Unified Identity, Information, and Scale-Coalescence Formulation

A Theory of Particle Identity, Information Propagation, and Emergent Structure Across Physical Scales

THEORETICAL DIVISION

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Abstract. This document formalises the Unified Identity, Information, and Scale-Coalescence Formulation (UIISCF): a theoretical framework that unifies classical molecular dynamics, subatomic particle identity, information accounting, and emergent structure under a single symbolic layer. The formulation is grounded in measured data from high-energy collider experiments and calibrated against the established hadron spectrum, nuclear binding energies, and atomic ionisation data. Four primitives organise the theory: the identity-bearing particle $\mathscr{P}_i$, the recoverable information Ξ_i, the interaction conductance $\mho_{ij}$, and the active scale-identity quota $\mathcal{Q}_S$. Two governing laws constrain all interactions and emergence events. Nineteen sections cover the complete theoretical and implementation specification, from global state representation through detector reconstruction, solver comparison, and version gating.

Contents

1. Motivation and Theoretical Basis

1.1 The Problem with Particle Identity

In orthodox quantum field theory, a particle is a quantised excitation of a field — an eigenstate of the particle-number operator. This description is mathematically complete but operationally opaque. At a particle collider, no particle is ever directly observed: the detector records energy deposits, timing signals, and ionisation trails. What experimenters call a "proton" or an "electron" is a reconstruction: a chain of inference from raw detector hits through track-fitting algorithms, calorimeter clustering, and vertex-finding to an assigned identity label and four-momentum. The particle, in this sense, is the output of an information-processing pipeline, not a primitive input to it.

This observation is not merely philosophical. The reconstruction efficiency, the mis-identification rate, the momentum resolution — all of these are information-theoretic quantities. When a π^+ and a π^- are produced in the same collision, the question of which track belongs to which pion is a question about recoverable information, not about the "true" identities of the particles in any absolute sense. The identities are exactly as distinguishable as the data permits them to be.

The Unified Identity, Information, and Scale-Coalescence Formulation takes this observation seriously and elevates it to a design principle. Identity is not a God-given label attached to a particle at creation; it is a quantity of distinguishability that is budgeted, transferred, and depleted as particles interact. The recoverable information Ξ_i is the formal expression of this budget.

1.2 Grounding in CERN Experimental Data

The scale ladder at the heart of UIISCF is not arbitrary. Each rung corresponds to a level of matter at which experimental data is available to constrain the parameters.

Scale 0 — Quarks and the Cornell potential. The confining potential between quarks, $E_{\text{conf}}(r) = \kappa r + \alpha_s / r$, was not invented for this formalism. It was extracted from quarkonium spectroscopy data: the J/ψ, Υ, and charmonium excited-state mass splittings measured at SLAC, CERN, and the Tevatron constrain $\kappa \approx 0.18\,\text{GeV}^2$ and $\alpha_s \approx 0.12$–0.30 at hadronic scales. The colour-averaged interaction (CAI) model used here is a direct approximation to the effective quark-antiquark potential inferred from these spectra, not a first-principles QCD calculation. The `full_qcd = false` flag is mandatory precisely because the distinction matters.

Scale 1 — Hadron formation and nuclear binding. The bonding threshold $\ell^{(1)}_{\text{bond}} \approx 10^{-4}$ Å and the birth criterion $m \geq 2$ (meson) or $m \geq 3$ (baryon) are set by the measured hadron sizes from electron–proton scattering at HERA and the proton charge radius from precision hydrogen spectroscopy. The baryon asymmetry in scale-1 coalescence — that three quarks form a baryon more readily than two — is a direct consequence of the colour-neutrality constraint on the triplet state.

Scale 2 — Atoms and classical MD. At the atomic scale the formalism reduces exactly to classical molecular dynamics. The force fields (Lennard-Jones, Morse, CHARMM, AMBER) used at this scale are calibrated against crystal structure data, neutron scattering spectra, and ab initio quantum chemical calculations. The classical MD bridge (Section 5) makes this reduction exact and not approximate: no information from the subatomic layers leaks into the classical layer unless `info_track` is explicitly enabled.

Scales 3–4 — Molecules and clusters. Bonding thresholds at the molecular and cluster scales are set from measured bond lengths (crystallographic databases), dissociation energies (gas-phase thermochemistry), and cluster stability data from mass spectrometry.

1.3 Why These Four Primitives

The choice of $\mathscr{P}_i$, Ξ_i, $\mho_{ij}$, and $\mathcal{Q}_S$ as the four primitives of the formalism is not arbitrary. Together they form a minimal complete basis for describing identity, information, coupling, and emergence.

The *identity-bearing particle* $\mathscr{P}_i$ is the carrier: it is the thing that has properties, occupies space, and participates in interactions. Without an identity carrier, there is nothing to track. The calligraphic $\mathscr{P}$ notation is chosen to distinguish it from the Hamiltonian operator $\hat{H}$ or the momentum p, with which it must coexist in the same equations.

The *recoverable information* Ξ_i quantifies how distinguishable $\mathscr{P}_i$ is from its environment. This is the quantity that decreases when particles interact (SM's First Law) and is partially inherited when new identities form (SM's Second Law). Using the Greek capital Ξ — a letter with no strong prior claim in this domain — deliberately avoids confusion with entropy S, energy E, or the wavefunction Ψ. The unit choice (nats) is deliberate: nats are the natural unit of information in continuous distributions, compatible with the thermodynamic entropy in the same units via $S = k_\mathrm{B}\Xi$.

The *interaction conductance* $\mho_{ij}$ encodes the rate at which information flows between $\mathscr{P}_i$ and $\mathscr{P}_j$ when they interact. The siemens-derived symbol $\mho$ is a deliberate analogy: just as electrical conductance governs charge flow between two nodes, interaction conductance governs information flow between two identities. A high-conductance pair exchanges information rapidly; a zero-conductance pair does not interact.

The *scale-identity quota* $\mathcal{Q}_S$ counts how many active identities exist at each scale. It is an integer quantity because identity is discrete: you cannot have half a proton. Its dynamics (SM's Second Law) describe emergence — the birth of higher-scale identities from lower-scale coalescence — in a way that is countable, verifiable, and conserved within the constraints of baryon number and lepton number.

1.4 A More Intuitive Representation of Matter

The standard approach to multi-scale simulation runs separate models at separate scales and couples them at boundaries: a quantum mechanics / molecular mechanics (QM/MM) boundary, or a coarse-grained / fine-grained boundary. These boundaries are artificial and generate artefacts: force discontinuities, energy drift, and ambiguity about which particles belong to which region.

UIISCF removes the explicit boundary by treating scale as a dynamic property of the identity, not a property of the simulation domain. A carbon atom is a scale-2 identity. If sufficient energy is deposited, it can be ionised (losing recoverable information to the environment) or, in extreme conditions, resolved into its nuclear constituents (a scale-1 event, only reachable with the subatomic channels active). The same state vector $\Omega(t)$ holds the atom regardless of which channels are active; the active channels determine which interactions are resolved and which are approximated.

This architecture mirrors how matter actually works: the nucleus is always there inside the atom, but at chemical energies its internal degrees of freedom are frozen out. UIISCF makes this freezing-out explicit through version gating and scale-channel weights, rather than hiding it inside a force field. The result is a representation of matter that can be understood from the bottom up, starting with quarks and building to clusters, rather than bolted together from separate codes that happen to share a boundary.

2. Notation Layer

Four primitive symbols are introduced. All subsequent formalism is expressed in terms of these primitives and the standard MD state. The motivation for each choice is discussed in Section 1; here the symbols are defined precisely.

Table 1: UIISCF primitive notation.

Symbol	LaTeX macro	Meaning	Units / Domain
$\mathscr{P}_i$	`\mathscr{P}_i`	Identity-bearing particle at scale n	Dimensionless label
Ξ_i	`\Xi_i`	Recoverable distinguishability (information) of $\mathscr{P}_i$	nats
$\mho_{ij}$	`\mho_{ij}`	Interaction conductance between $\mathscr{P}_i$ and $\mathscr{P}_j$	[nats fs^{-1}]
$\mathcal{Q}_S$	`\mathcal{Q}_S`	Active scale-identity quota at scale S	$\mathbb{Z}_{\geq 0}$

The symbol n denotes scale level: $n = 0$ (quark), $n = 1$ (hadron/nucleus), $n = 2$ (atom), $n = 3$ (molecule), and so on. Scale levels are open-ended; an implementation gates which levels are active via version constraints (Section 20).

The two governing laws are stated here in full and recur as interjections throughout the formalism wherever they constrain a derivation.

First Law of Identity — Conservation of Distinguishability.

Every pairwise interaction redistributes or consumes recoverable information. No interaction is information-neutral. Formally:

$$\mathscr{P}_i \leftrightarrow \mathscr{P}_j \implies \Delta\Xi_{ij} < 0 \text{ or } \Delta S_{ij} > 0,$$

where $\Delta\Xi_{ij} = \Xi_i' + \Xi_j' - \Xi_i - \Xi_j$ is the change in total recoverable information and ΔS_{ij} is the entropy produced by the interaction. Equality $\Delta\Xi_{ij} = 0$, $\Delta S_{ij} = 0$ is forbidden for any physical interaction.

Second Law of Identity — Scale-Coalescence Birth Criterion.

When $m \geq 3$ lower-scale identities satisfy mutual proximity and bond maturity, a new higher-scale identity is born:

$$m \geq 3, \quad d_{ij}^{(n)} \leq \ell_{\text{bond}}^{(n)}, \quad B_n(C_n) \geq \tau_B \implies \text{Exist}\left(\mathscr{P}_*^{(n+1)}\right) = 1, \quad \Delta\mathcal{Q}_S = +1,$$

where $d_{ij}^{(n)}$ is the pairwise separation at scale n, $\ell_{\text{bond}}^{(n)}$ is the scale-n bonding length threshold, $B_n(C_n)$ is the bond maturity functional evaluated on bond count C_n, and τ_B is the maturity threshold. The new identity $\mathscr{P}_*^{(n+1)}$ inherits summed mass, net charge, and a fresh Ξ budget.

3. Global State

The simulation state at time t is the tuple

$$\Omega(t) = (\mathbf{R}(t),\ \mathbf{V}(t),\ \mathbf{F}(t),\ \mathbf{M},\ \mathbf{Z},\ \boldsymbol{\Xi}(t),\ \boldsymbol{\mho}(t),\ \mathcal{Q}_S(t),\ \mathbf{H}(t),\ E_{\text{tot}}(t),\ t,\ \Delta t),$$

where $\mathbf{R}$, $\mathbf{V}$, $\mathbf{F} \in \mathbb{R}^{N\times 3}$ are position, velocity, and force arrays over N particles; $\mathbf{M} \in \mathbb{R}^N$ is the mass vector; $\mathbf{Z} \in \mathbb{Z}^N$ is the atomic-number / particle-species vector; $\boldsymbol{\Xi}(t) \in \mathbb{R}^N$ is the per-particle recoverable information vector; $\boldsymbol{\mho}(t)$ is the sparse conductance matrix (non-zero only for bonded or proximate pairs); $\mathcal{Q}_S(t) \in \mathbb{Z}_{\geq 0}$ is the active identity count at each scale; $\mathbf{H}(t)$ is the 3×3 cell matrix; E_{tot} is total energy; and Δt is the current timestep.
Compared to classical MD, the state Ω carries two new blocks: $\boldsymbol{\Xi}$ and $\boldsymbol{\mho}$. These blocks are dormant (zero-cost) when information tracking is disabled and active only when explicitly requested. This design ensures full backward compatibility with existing MD workflows: a simulation that never writes to these blocks behaves identically to a classical MD run.

3.1 State-Vector Projection

The universal state vector $\mathbf{Q}$ projects onto Ω via:

$$\mathbf{Q}(t) \;=\; \big[\, \mathbf{R}^\top \mid \mathbf{V}^\top \mid \boldsymbol{\Xi}^\top \mid \operatorname{vec}(\boldsymbol{\mho}) \,\big]^\top.$$

Every observable (pressure, temperature, spectra, identity counts) is a projection $\mathcal{P}_\alpha(\mathbf{Q})$ for some projection operator $\mathcal{P}_\alpha$. The identity-information block $[\boldsymbol{\Xi}^\top \mid \operatorname{vec}(\boldsymbol{\mho})]$ is new in UIISCF and was absent from the classical MD projection.

3.2 Initialisation Invariants

At $t = 0$ the following must hold:

1. $\Xi_i(0) = \Xi_i^{\text{birth}}$ for every particle, set from the species registry.
2. $\mho_{ij}(0) = 0$ for all pairs unless an explicit bond is declared in the input block.
3. $\mathcal{Q}_S(0) = N$ (all particles are scale-n identities).
4. $\mathbf{H}(0)$ is orthonormal and volume-positive: $\det(\mathbf{H}) > 0$.

4. Scale Identity Ladder

The scale ladder is the structural backbone of UIISCF. Rather than committing to a single level of description, the formalism assigns each identity to a rung n and allows transitions between rungs through coalescence (upward) and dissociation (downward). The parameters at each rung are not freely adjustable: they are anchored to measured data. The bonding lengths come from scattering cross-sections and structural databases; the birth thresholds come from nuclear and molecular stability measurements; the maturity exponents are calibrated so that the formation rate of common hadrons and small molecules reproduces known reaction rates at the relevant temperatures.

The reason for requiring $m \geq 3$ as the minimum coalescence count (rather than two, as in a naive bonding model) is both physical and computational. Physically, two-body encounters rarely produce stable bound states without a third body to carry away excess momentum: this is the standard three-body recombination argument in atomic and nuclear physics. Computationally, requiring a minimum of three participants prevents spurious coalescence events from cascading through the identity ladder during periods of high-density fluctuation.

Particles are organised on a discrete scale ladder. Each rung n has its own bonding threshold, maturity functional, and information budget.

Table 2: Scale identity ladder rungs. Thresholds are calibrated from experimental data as described in Section 1.

Scale n	Entity type	$\ell^{(n)}_{\text{bond}}$ (Å)	τ_B	Birth criterion (min. m)
0	Quark	10^{-5}	0.90	3 (colour-neutral triplet)
1	Hadron / nucleus	10^{-4}	0.85	2 (meson) or 3 (baryon)
2	Atom	2.5	0.70	2
3	Molecule	5.0	0.65	3
4	Cluster	15.0	0.60	5

4.1 Bond Maturity Functional

The bond maturity functional is chosen to be a saturating exponential:

$$B_n(C_n) \;=\; 1 - \exp(-\lambda_n C_n)\,,$$

where C_n is the cumulative bond-event count at scale n and $\lambda_n > 0$ is the maturity rate parameter. This functional form is chosen deliberately. A linear maturity criterion would allow coalescence to trigger on the first interaction, which is too aggressive. A hard threshold on C_n would produce discontinuous birth rates. The saturating exponential is the simplest functional that is continuous, monotonically increasing, bounded in $[0, 1)$, and parameterised by a single rate constant whose physical interpretation is straightforward: $1/\lambda_n$ is the characteristic interaction count for maturation. Default values: $\lambda_0 = 5,\ \lambda_1 = 3,\ \lambda_2 = 1.5,\ \lambda_3 = 0.8$.

4.2 Information Budget at Birth

When the Second Law of Identity fires and $\mathscr{P}^{(n+1)}_*$ is born:

$$\Xi^{(n+1)}_* \;=\; \eta \sum_{i\in\mathcal{C}} \Xi^{(n)}_i \;-\; \Delta S_{\text{coal}},$$

where $\mathcal{C}$ is the coalescing set, $\eta \in (0, 1]$ is the information-transfer efficiency (default $\eta = 0.95$), and $\Delta S_{\text{coal}} > 0$ is the coalescence entropy cost. The deficit $\sum_i \Xi_i - \Xi_*/\eta$ is written to the entropy ledger.

First Law of Identity applies at coalescence: the information loss $\sum_i \Xi^{(n)}_i - \Xi^{(n+1)}_* > 0$ must be strictly positive. A coalescence that conserves or increases total recoverable information is rejected with error `ERR_COAL_INFO_NONZERO`.

5. Classical MD Bridge

For scale-2 (atomic) identities the UIISCF reduces exactly to standard molecular dynamics. This is not a coincidence or a special case; it is an explicit design requirement. The classical MD literature represents decades of validated force fields, thermostat algorithms, barostat algorithms, and long-range solvers. Any formalism that introduces a subatomic layer while breaking the classical layer loses the trust of the community that already uses classical MD. The bridge is therefore designed so that activating information tracking adds exactly two new ODEs per particle ($\dot{\Xi}_i$ and $\dot{\mho}_{ij}$) without modifying the existing force, position, or velocity equations.

5.1 Equations of Motion

$$m_i \ddot{\mathbf{r}}_i \;=\; \mathbf{F}_i \;=\; -\nabla_{\mathbf{r}_i} U,$$
$$U \;=\; \sum_{i<j} \phi(r_{ij}) \;+\; \sum_{\text{bonds}} V_{\text{bond}} \;+\; \sum_{\text{angles}} V_{\text{angle}} \;+\; \sum_{\text{dihedrals}} V_{\text{dihed}}.$$

Periodic boundary conditions apply via the minimum-image convention using cell matrix $\mathbf{H}$:

$$\mathbf{r}_{ij}^{\text{MIC}} \;=\; \mathbf{H}(\mathbf{s}_{ij} - \text{round}(\mathbf{s}_{ij}))\,, \quad \mathbf{s}_{ij} = \mathbf{H}^{-1}(\mathbf{r}_i - \mathbf{r}_j).$$

5.2 Information Coupling to MD

At scale 2 the recoverable information Ξ_i is a passive bookkeeping quantity computed alongside positions. For every force evaluation step:

$$\dot{\Xi}_i \;=\; -\sum_{j\neq i} \mho_{ij}\,(\Xi_i - \Xi_j) \;-\; \gamma_i \Xi_i,$$

where $\gamma_i \geq 0$ is a species-specific information dissipation rate and the conductance $\mho_{ij}$ is proportional to the pair interaction strength. This equation has the structure of a heat diffusion equation on the interaction graph: information flows from high-Ξ particles to low-Ξ particles along edges weighted by conductance, and dissipates at rate γ_i. The analogy with heat conduction is intentional and makes the physical interpretation transparent.

```
[run]
integrator   = "velocity_verlet"
dt           = 0.5          # fs
steps        = 200000
thermostat   = "nose_hoover"
barostat     = "parrinello_rahman"
info_track   = true         # enable Xi / mho bookkeeping
info_dissip  = 0.001        # gamma_i default (fs^-1)
```

Listing 1: Classical MD configuration block (reference scripting format).

6. Matrix-Force Policy

Classical MD codes hard-code force-computation loops: one loop for pair potentials, one for bonds, one for angles, and so on. Adding a subatomic force channel in this paradigm requires modifying the innermost loops, which risks breaking validated classical behaviour and makes it difficult to gate subatomic contributions by scale or version.

The matrix-force policy solves this by routing all force computation through a weighted dispatch table. Each force channel contributes a force vector $\mathbf{F}_i^{(\alpha)}$; the total force is a weighted sum. Channel weights are set to zero by default for all subatomic channels and raised to 1.0 only when the corresponding scale is active. This architecture means that adding a new channel — colour-averaged interaction, gluon proxy, weak proxy — never modifies an existing channel, and can always be validated in isolation before being combined. The classical path is preserved exactly: with all subatomic weights at zero, the dispatch table produces the identical result to the original hard-coded loops.

6.1 Force Dispatch Table

$$\mathbf{F}_i \;=\; \sum_{\alpha} w_\alpha \, \mathbf{F}_i^{(\alpha)},$$

where α indexes active force channels and w_α is a channel weight gated by the version and active-scale flags.

Table 3: Force channel dispatch table.

Channel α	Description	Default w_α	Min. version
`pair`	Pair potential (LJ, Morse, . . .)	1.0	v5.1.4
`bond`	Harmonic/FENE bond	1.0	v5.1.4
`angle`	Valence angle	1.0	v5.1.4
`dihedral`	Dihedral torsion	1.0	v5.1.4
`coulomb`	Long-range Coulomb (PPPM)	1.0	v5.1.4
`cai`	Colour-averaged interaction (subatomic)	0.0	v5.1.5
`gluon`	Cornell proxy (effective QCD)	0.0	v5.1.5

6.2 Policy Enforcement

The engine validates the weight vector before every run:

1. $\sum_\alpha w_\alpha > 0$ (at least one active channel).
2. Subatomic channels (`cai`, `gluon`) require `subatomic_mode = true` and kernel $\geq$ v5.1.5.
3. If `gluon` weight > 0 then `full_qcd = false` *must* be explicitly set. Omission raises `ERR_QCD_FLAG_MISSING`.

First Law of Identity constrains channel coupling: enabling a new force channel always changes Ξ accounting. Adding `cai` or `gluon` without enabling `info_track` is forbidden; the implementation will reject the configuration with `ERR_INFO_TRACK_REQUIRED`.

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7. Per-Atom Event-Energy State

Continuous force integration handles the smooth evolution of positions and velocities but is not the right tool for discontinuous events: bond formation, bond breakage, coalescence, annihilation. These events change the topology of the interaction graph instantaneously and deposit or release discrete amounts of energy. Tracking them requires a dedicated event-energy register per particle, separate from the running kinetic and potential energy.

Separating event energy from kinetic and potential energy also provides a natural diagnostic: the event-energy trace is a record of every discrete transition that occurred, in time order, with position, energy, and information-change metadata. This trace can be replayed, audited, and compared across solver runs independently of the continuous trajectory.

Each particle carries:

$$\mathcal{E}_i(t) \;=\; \big(E_i^{\mathrm{kin}},\; E_i^{\mathrm{pot}},\; E_i^{\mathrm{event}},\; \Xi_i,\; n_i^{\mathrm{evt}}\big),$$

where E_i^{event} is the cumulative energy from discrete events and n_i^{evt} is the event counter.

7.1 Event Types

Table 4: Discrete event types and their energy/information effects.

Event type	Code	ΔE^{event}	$\Delta\Xi$	$\Delta\mathcal{Q}_S$
Bond formation	`BOND_FORM`	$-E_{\text{bond}}$	< 0	0
Bond break	`BOND_BREAK`	$+E_{\text{bond}}$	< 0	0
Coalescence	`COAL`	$-\Delta E_{\text{coal}}$	$\leq -\Delta S_{\text{coal}}$	$+1$
Dissociation	`DISSOC`	$+\Delta E_{\text{dissoc}}$	< 0	-1
Annihilation	`ANNIH`	$+2m_i c^2$	$-\Xi_i - \Xi_{\bar{i}}$	-2
Excitation	`EXCITE`	$+E_\gamma$	< 0	0
De-excitation	`DEEXCITE`	$-E_\gamma$	< 0	0

7.2 Event Log Format

Each event is appended to the event log (`.evlog` artifact) as:

```
STEP   TYPE         ID_A   ID_B   SCALE  dE_eV          dXi
12500  BOND_FORM    1042   2107   2      -4.32100e+00   -0.0023
12501  COAL         0133   0134   1      -1.20000e+02   -1.4400
```

8. Event Limiter

Unconstrained event rates can destabilise numerical integration. A single timestep that generates hundreds of coalescence events deposits a large impulsive energy into the system, causing temperature spikes and, in extreme cases, the velocity-Verlet integrator to step a particle through a potential barrier. The event limiter is a numerical stability mechanism, not a physical one: it does not claim that real matter has a maximum event rate, only that the discrete-time integrator does.

The rate ceiling is set proportional to $N \cdot \Delta t$ so that the ceiling scales appropriately with system size and timestep, avoiding the pathological situation where a fine-grained timestep unexpectedly allows more events per step than a coarse one.

8.1 Rate Ceiling

$$n_{\text{evt}}^{\text{step}} \;\leq\; \lfloor \lambda_{\text{evt}} \cdot N \cdot \Delta t \rfloor \,,$$

where λ_{evt} is the maximum event rate density (events per particle per fs), default $\lambda_{\text{evt}} = 0.01$. Events exceeding the ceiling in a single step are deferred to the next step via a FIFO queue.

8.2 Energy-Change Guard

For any single event:

$$\left|\Delta E_i^{\text{event}}\right| \;\leq\; \alpha_{\text{guard}} \cdot E_i^{\text{kin}},$$

where $\alpha_{\text{guard}} = 5.0$ by default. Events violating this guard are logged as `GUARD_CLIP` and the energy change is clamped.

First Law of Identity applies to clamped events: even a clipped energy change must satisfy $\Delta\Xi < 0$. A clamp that would produce $\Delta\Xi \geq 0$ is rejected entirely and logged as `ERR_GUARD_INFO_VIOLATION`.

```
[kernel]
event_rate_density  = 0.01    # events per particle per fs
event_energy_guard  = 5.0     # alpha_guard
event_queue_depth   = 1000    # max deferred events
event_log_path      = "sim.evlog"
```

Listing 2: Event limiter configuration block.

9. Formation-Defect Bridge

Coalescence and dissociation leave traces in the trajectory record beyond the continuous position and velocity data. The formation-defect bridge is the mechanism by which these traces are captured in a dedicated formation trajectory format and linked back to the identity ladder.

The information-loss fraction δ_{defect} at dissociation provides a physically meaningful severity classification that has no analogue in purely energetic defect classification schemes. A clean split ($\delta < 0.02$) indicates that the two fragments carry nearly the same distinguishability they had before binding; the information was stored in the interaction, not consumed. A fragmentation event ($\delta > 0.1$) indicates that most of the identity information of the parent was lost to the environment — the fragments are fundamentally different objects, not simply pieces of the original.

The formation trajectory connects formation events to coalescence transitions on the identity ladder. Defect events connect to dissociation and bond-break transitions.

9.1 Formation Event Record

A formation event is triggered when the Second Law of Identity fires:

```
# formation event at step 50000
FORM  parent_ids=[1042,1043,1044]  child_id=5000  scale=2->3
      mass_sum=28.054 charge_sum=0  Xi_child=2.8500
      dS_coal=0.0247  bond_maturity=0.9200
```

9.2 Defect Classification

Defects are classified by the information change at dissociation:

$$\delta_{\mathrm{defect}} = \frac{\Xi_{\mathrm{parent}} - \sum_k \Xi_k^{\mathrm{fragment}}}{\Xi_{\mathrm{parent}}}.$$

Table 5: Defect classification by information loss fraction.

Class	δ_{defect} range	Label	Severity
Clean split	$[0, 0.02)$	`DEFECT_CLEAN`	Low
Lossy split	$[0.02, 0.1)$	`DEFECT_LOSSY`	Medium
Fragmentation	$[0.1, 1.0]$	`DEFECT_FRAG`	High

Second Law of Identity sets the lower bound on coalescence count: $m \geq 3$. Any two-body coalescence at scale $n \geq 2$ is promoted to a `COAL_PROBE` event and only finalised if a third proximate particle joins within $\tau_{\text{probe}} = 5\,\text{fs}$ (configurable).

10. Quark Component Matrix

The quark sector is the point at which this formalism is most directly accountable to measured data. The parameters that appear here are not fitted to a simulation objective; they are inherited from experimental hadron physics. The Cornell potential string tension $\kappa \approx 0.18\,\text{GeV}^2$ is the value inferred from the linear Regge trajectories of light-quark mesons as measured at CERN's NA series and the Fermilab fixed- target experiments. The running coupling $\alpha_s \approx 0.18$ at the relevant hadronic scale is consistent with the world average from the PDG compilation of $\alpha_s(m_Z) = 0.1179 \pm 0.0010$ evolved down to $Q \sim 1\,\text{GeV}$ using two-loop running. The hyperfine coupling V_{hyp} is calibrated from the J/ψ–η_c and Υ–η_b mass splittings.

Working backwards from CERN collision data to a tractable simulation model reveals something useful: quark identity is less a property of the quark itself than of the hadron it inhabits. A quark confined in a proton is a different object — in terms of its effective mass, its coupling, and its contribution to conserved quantum numbers — from a quark in a pion or in the deconfined quark-gluon plasma produced in heavy-ion collisions at the LHC. The quark component matrix encodes this context-dependence explicitly through the recoverable information column $\Xi_i^{(0)}$, which varies between hadronic environments.

At scale $n = 0$ each quark is an identity $\mathscr{P}_i^{(0)}$ carrying colour charge $c_i \in \{R, G, B, \bar{R}, \bar{G}, \bar{B}\}$, flavour f_i, and spin $\sigma_i = \pm\frac{1}{2}$. The quark component matrix $\mathbf{Q}^{(0)}$ has one row per quark:

$$Q_{ik}^{(0)} = \begin{cases} m_i/m_u & k = 1 \quad \text{(mass in units of } m_u\text{)} \\ q_i & k = 2 \quad \text{(electric charge)} \\ c_i^{\text{int}} & k = 3 \quad \text{(colour integer: } 1 \ldots 6\text{)} \\ f_i^{\text{int}} & k = 4 \quad \text{(flavour: } u = 1, d = 2, s = 3, c = 4, b = 5, t = 6\text{)} \\ \sigma_i & k = 5 \quad \text{(spin: } \pm 1\text{)} \\ \Xi_i^{(0)} & k = 6 \quad \text{(recoverable information)} \end{cases}$$

10.1 Colour-Neutral Hadron Formation

Quarks form hadrons when a colour-neutral triplet (or pair for mesons) satisfies the Second Law of Identity at scale 0. The colour-neutrality condition:

$$\bigoplus_{i \in \mathcal{C}} c_i = \text{white},$$

where $\oplus$ denotes colour addition. Colour neutrality is checked before applying the birth criterion.

10.2 Colour-Averaged Interaction (CAI)

Because full QCD is not implemented, a colour-averaged interaction (CAI) channel approximates the strong force between quarks. The CAI potential:

$$V_{\text{CAI}}(r_{ij}) = -\frac{\alpha_s}{r_{ij}}\,\delta_{c_i\bar{c}_j} + \kappa_{\text{str}} r_{ij} + V_{\text{hyp}},$$

where $\alpha_s \approx 0.12$–0.30, $\kappa_{\rm str} \approx 0.18\,\mathrm{GeV}^2$ (Cornell string tension), and $V_{\rm hyp}$ is the hyperfine correction from spin-colour coupling.

```
[quark]
enabled        = true
flavours       = ["u","d","s"]   # active flavour set
alpha_s        = 0.18
kappa_string   = 0.18            # GeV^2
hyp_coupling   = 0.02
full_qcd       = false           # MANDATORY: always set false
cai_channel    = true
info_track     = true
```

Listing 3: Quark configuration block (reference implementation).

11. Lepton Identity Matrix

Leptons are fundamental-scale identities that do not participate in the strong interaction. They are among the most precisely measured particles in the Standard Model: the electron mass is known to 13 significant figures, the muon anomalous magnetic moment to 10. This precision makes the lepton sector an unusually clean test of identity accounting: the recoverable information of an electron is its electric charge, mass, and spin — three quantities that never change in any interaction short of annihilation. Ξ_e is therefore essentially constant until the electron annihilates, at which point it drops to zero discontinuously. This is the clearest example of the First Law of Identity at work: no interaction changes the electron's Ξ until the one interaction that destroys it entirely.

Each lepton $\mathscr{P}_i^{(\ell)}$ occupies one row of the lepton identity matrix $\mathbf{L}$:

$$L_{ik} = \begin{cases} m_i/m_e & k=1 \quad \text{(mass in electron masses)} \\ q_i & k=2 \quad \text{(charge: } -1, 0) \\ g_i & k=3 \quad \text{(generation: 1, 2, 3)} \\ \sigma_i & k=4 \quad \text{(spin)} \\ L_i^{\rm num} & k=5 \quad \text{(lepton number: } \pm 1) \\ \Xi_i^{(\ell)} & k=6 \quad \text{(recoverable information)} \end{cases}$$

11.1 Lepton Interactions

Leptons interact via three channels in the reference implementation:

1. **Electromagnetic:** Coulomb interaction with all charged particles via the PPPM long-range solver.
2. **Weak proxy:** A Yukawa-type screened potential approximating W/Z-boson exchange (range $\sim 10^{-3}$ Å, gated by `weak_proxy = true`).
3. **Electron shell binding:** At scale 2, electrons are not simulated as explicit particles by default; their effect enters via the force field. Explicit electron tracking requires `explicit_electrons = true` (v5.1.5+).

First Law of Identity is especially active for lepton-antilepton encounters. An electron meeting a positron undergoes annihilation (`ANNIH`), releasing $\Delta\Xi = -\Xi_e - \Xi_{\bar{e}}$ and energy $2m_e c^2 + E_{\text{kin}}$. The information is written to the entropy ledger irreversibly.

12. Boson Identity Matrix

Gauge bosons occupy an unusual position in the UIISCF framework. In the Standard Model they are the carriers of forces; in the simulation they are either handled analytically (photons, via potential energy) or approximated by proxy potentials (gluons, via the Cornell model; W/Z, via Yukawa screening). The decision to represent bosons as transient matrix entries rather than persistent identities reflects a deliberate modelling choice: at the timescales and length scales of the simulation, an exchanged virtual photon has a lifetime many orders of magnitude shorter than the timestep. Treating it as a persistent particle would be physically inaccurate and computationally wasteful.

The boson identity matrix is therefore a log of mediator events, not a roster of persistent particles.

Gauge bosons are mediator particles represented as transient entries in the boson identity matrix $\mathbf{B}$:

$$B_{ik} = \begin{cases} m_i^{\text{eff}} & k=1 \quad \text{(effective mass in GeV}/c^2\text{)} \\ q_i & k=2 \quad \text{(charge)} \\ s_i & k=3 \quad \text{(spin: 0, 1, 2)} \\ \tau_i^{\text{life}} & k=4 \quad \text{(lifetime in fs, or } \infty\text{)} \\ \text{type}_i & k=5 \quad \text{(photon, } W^{\pm}\text{, } Z^0\text{, gluon, Higgs proxy)} \\ \Xi_i^{(b)} & k=6 \quad \text{(near zero for gauge bosons)} \end{cases}$$

12.1 Gluon Proxy Model

The gluon proxy is an effective model. Full QCD colour dynamics are not implemented. The Cornell potential governs quark confinement in place of explicit gluon exchange:

$$E_{\text{conf}}(r) = \kappa r + \frac{\alpha_s}{r}, \quad \kappa \approx 0.18\,\text{GeV}^2, \quad \alpha_s \approx 0.1\text{–}0.3.$$

This is a Cornell string-plus-Coulomb potential calibrated to reproduce hadron mass splittings at low energy. Its parameters are taken directly from quarkonium spectroscopy, as described in Section 10. It is *not* a perturbative QCD calculation.

Mandatory disclaimer: Any configuration using quark or boson channels must explicitly set `full_qcd = false`. Omission raises `ERR_QCD_FLAG_MISSING` at parse time. This requirement exists to prevent Cornell-proxy results from being cited as genuine QCD predictions.

12.2 Photon Emission Proxy

Photon emission (de-excitation) is handled as an event in the event log, not as a propagating particle. The emitted energy is $E_\gamma = \Delta E$, recorded as `DEEXCITE`. Photon absorption is the time-reverse.

13. Relation-Particle Matrix

Classical MD tracks particles and their properties but rarely tracks the history of individual interactions in structured form. The relation-particle matrix fills this gap by recording, for every currently active particle pair, the current conductance, separation, information history, and bond type. This matrix is the information substrate on which the First Law operates: the $\Delta\Xi_{ij}^{\text{last}}$ column is a direct record of information transferred at each event, and $\Delta S_{ij}^{\text{cum}}$ is the running entropy produced by the pair.

The relation-particle matrix $\mathbf{R}_{\text{rel}}$ encodes pairwise relationships. Each row is a directed pair (i, j):

$$(R_{\text{rel}})_{(ij),k} \;=\; \begin{cases} \mho_{ij} & k=1 \quad \text{(interaction conductance)} \\ d_{ij} & k=2 \quad \text{(current separation)} \\ \Delta\Xi_{ij}^{\text{last}} & k=3 \quad \text{(information delta at last event)} \\ \Delta S_{ij}^{\text{cum}} & k=4 \quad \text{(cumulative entropy from pair)} \\ t_{ij}^{\text{last}} & k=5 \quad \text{(time of last event)} \\ \text{bond}_{ij} & k=6 \quad \text{(bond type: 0=none, 1=covalent, 2=ionic,\dots)} \end{cases}$$

13.1 Conductance Dynamics

The conductance $\mho_{ij}$ evolves between force steps:

$$\dot{\mho}_{ij} \;=\; \alpha_{\mho}\big(\mho_{ij}^{\text{eq}}(r_{ij}) - \mho_{ij}\big) \;-\; \beta_{\mho}\mho_{ij},$$

where $\mho_{ij}^{\text{eq}}(r)$ is the equilibrium conductance (decreasing with r, zero beyond r_{cut}), $\alpha_{\mho}$ is the relaxation rate, and $\beta_{\mho}$ is the natural decay rate.

14. Proper-Time Identity Matrix

At sub-relativistic velocities ($v \ll c$) proper time coincides with simulation time and the proper-time identity matrix is diagonal: $T_{ii} = t$. The full relativistic extension is anticipated for a future release and is sketched here for completeness.

14.1 Lorentz Factor Correction

For a particle with velocity v_i:

$$\tau_i(t) \;=\; \int_0^t \sqrt{1 - \frac{v_i^2(t')}{c^2}}\, dt' \;\approx\; t\left(1 - \frac{\langle v_i^2 \rangle}{2c^2}\right).$$

The proper-time lag $T_{ij} = \tau_i(t) - \tau_j(t)$ for $i \neq j$ becomes significant in beam-simulation scenarios where particles are accelerated to relativistic energies: two quarks produced in the same collision but boosted in opposite directions accumulate different proper times and therefore age their identity information at different rates.

14.2 Information Age

The proper-time formalism provides a natural measure of information age:

$$\Xi_i^{\text{age}}(\tau) \;=\; \Xi_i(0)\exp(-\kappa_{\text{age}}\tau_i),$$

where κ_{age} is an information half-life parameter, relevant for long-lived radioactive species where the identity information of the parent nucleus degrades as it approaches decay.

First Law of Identity in the relativistic frame: the interaction must satisfy $\Delta\Xi_{ij} < 0$ or $\Delta S_{ij} > 0$ in the lab frame. Lorentz boosts do not exempt an interaction from the law; information accounting is performed in the lab frame before any boost is applied.

---

15. Dark-Sector Projection

The dark-sector projection is included in the UIISCF not because the nature of dark matter is understood, but precisely because it is not. The information-theoretic framework provides a natural language for dark matter: a dark-sector particle $\mathscr{P}_i^{(\text{dk})}$ is an identity that carries mass and therefore participates in gravitational clustering, but carries zero electromagnetic charge and therefore contributes nothing to detector hits. Its recoverable information is sequestered: it cannot be transferred to standard-model particles through the channels available at collider energies.

This sequestration is the formal expression of why dark matter is dark. It is not absent from the simulation; it is present but its Ξ is not accessible to any standard-model detector projection operator.

Dark-sector particles carry mass but do not participate in electromagnetic or strong interactions:

$$D_{ik} = \begin{cases} m_i^{\text{dk}} & k = 1 \\ 0 & k = 2 \quad \text{(no EM charge)} \\ \sigma_i^{\text{dk}} & k = 3 \\ \xi_i^{\text{dk}} & k = 4 \quad \text{(dark coupling)} \\ \Xi_i^{(\text{dk})} & k = 5 \end{cases}$$

15.1 Dark-Sector Force Channel

The dark-sector force is modelled as a Yukawa potential:

$$V_{\text{dk}}(r_{ij}) = -\xi_i \xi_j \frac{e^{-m_{\text{med}} r_{ij}}}{r_{ij}},$$

where m_{med} is the dark mediator mass. In the limit $m_{\text{med}} \to 0$ this reduces to a gravitational-like $1/r$ potential. The projection is explicitly *phenomenological*: no claim is made regarding correspondence with any particular dark matter model.

```
[dark_sector]
enabled        = false         # off by default
mediator_mass  = 1.0e-4       # GeV/c^2
coupling       = 0.001
info_track     = true
full_qcd       = false
```

Listing 4: Dark-sector configuration block.

16. Event/Annihilation Matrix

Annihilation is the most dramatic event in the UIISCF framework: two identities are extinguished and their rest mass converted to radiation. From an information-theoretic perspective, annihilation is the maximum possible violation of identity continuity. The incoming particles have well-defined identity ($\Xi_e, \Xi_{\bar{e}} > 0$); the outgoing photons carry no recoverable identity information ($\Xi_\gamma \approx 0$). The entire information budget of two particles is written to the entropy ledger in a single step.

The annihilation matrix records this event comprehensively. The record is designed to be auditable after the fact: given the annihilation matrix, one can verify that lepton number was conserved (one electron and one positron entered), that energy-momentum was conserved (the photon energies sum to $2m_ec^2+E_{\text{kin}}$), and that the entropy increase matches the total information destroyed.

$$A_{(ij),k} = \begin{cases} m_i + m_j & k=1 \quad \text{(total annihilated mass)} \\ 2(m_i+m_j)c^2 + E_{\text{kin}} & k=2 \quad \text{(energy released, eV)} \\ t_{\text{annihil}} & k=3 \quad \text{(simulation time)} \\ \mathbf{r}_{\text{annihil}} & k=4 \quad \text{(position)} \\ \Xi_i + \Xi_j & k=5 \quad \text{(total information destroyed)} \\ \Delta S_{\text{annihil}} & k=6 \quad \text{(entropy produced)} \end{cases}$$

16.1 Annihilation Guard Conditions

Annihilation is triggered when:

1. The pair (i, j) are confirmed particle-antiparticle.
2. $d_{ij} \le r_{\text{annihil}}$ (default 10^{-4} Å for leptons, 10^{-5} Å for hadrons).
3. $E_{\text{kin}}^{\text{rel}} \le E_{\text{annihil}}^{\text{max}}$.

First Law of Identity at annihilation: the information destroyed, $\Xi_i + \Xi_j$, is entirely unrecoverable. The engine writes the full Ξ budget to the entropy ledger and increments the global annihilation entropy counter $S_{\text{annihil}}^{\text{glob}}$. Annihilation without entropy accounting raises `ERR_ANNIHIL_INFO_MISSING`.

17. Detector Projection and Reconstruction

The detector projection module is the formal bridge between simulation and experiment. It mirrors the actual reconstruction pipeline used at particle colliders: raw detector hits are processed through clustering, track-seeding, momentum fitting, and vertex-finding to produce a list of reconstructed particles with four-momenta and identity assignments.

In a real experiment this pipeline runs in the opposite direction to the simulation: measured hits are the inputs and reconstructed particles are the outputs. Here the pipeline runs forward: true particle states are the inputs and synthetic hits are generated, then processed by the same reconstruction algorithm. This allows direct comparison between truth-level identity (from $\Omega(t)$) and reconstructed identity ($\tilde{\mathscr{P}}$), which is the key figure of merit for the identity formalism: if the reconstruction fails to separate two particles that the formalism treats as distinguishable, their Ξ values are too close and should be revised.

The detector module therefore acts as a continuous test of the information accounting: it tells you whether the Ξ values you have assigned to your particles are experimentally meaningful or not.

17.1 Projection Operator

A detector D_α with acceptance function $A_\alpha(\mathscr{P}_i)$ produces the hit vector:

$$\mathbf{h}^{(\alpha)}(t) \;=\; \sum_i A_\alpha(\mathscr{P}_i)\, \boldsymbol{\delta}(\mathbf{r}_i - \mathbf{r}_{D_\alpha}) \otimes \mathcal{E}_i(t),$$

where $\otimes$ denotes the detector response convolution.

17.2 Reconstruction Algorithm

1. **Hit clustering:** Group hits within spatial resolution δx_{det} and time window δt_{det}.
2. **Track seeding:** Each cluster with $n_{\mathrm{hit}} \geq 3$ seeds a candidate track.
3. **Kalman filter:** Propagate track hypothesis through the material budget and fit momentum.
4. **Vertex finding:** Intersect compatible tracks at the primary vertex; record secondary vertices for decay topologies.
5. **Identity assignment:** Assign reconstructed identity $\tilde{\mathscr{P}}$ and compare to truth-level $\mathscr{P}$ from simulation state.

```
# detector reconstruction output (.reco)
STEP   TRACK_ID   PX_GEV  PY_GEV  PZ_GEV  MASS_GEV  PID_TRUE  PID_RECO
50000  0001       0.4523  0.1204  2.3311   0.1396    211       211
50000  0002      -0.3901  0.0891  1.9823   0.1396   -211      -211
```

18. Solver Comparison State

The solver comparison state introduces a new dimension to numerical validation. Classical MD validation typically checks energy conservation and trajectory divergence between different integrators. UIISCF adds a third metric: information divergence $\delta\Xi_{\alpha\beta}$.

The significance of this metric is that it reveals a class of solver pathology that energy conservation alone cannot detect. A non-symplectic integrator (Runge-Kutta 4, Adams-Bashforth) can conserve energy to high precision on short runs while silently misattributing information flow between particle pairs. The $\delta\Xi$ metric catches this because the information-flow equation is sensitive to the order of force evaluation in a way that the energy integral is not. Symplectic integrators (velocity Verlet, PEFRL, Yoshida 6) preserve the information budget automatically because they preserve the symplectic structure of the phase space; non-symplectic integrators do not.

Two or more solver threads share the same initial state $\Omega(0)$ and evolve independently. Comparison at checkpoints:

$$\delta\Xi_{\alpha\beta}(t_c) \;=\; \left| \sum_i \Xi_i^{(\alpha)}(t_c) - \sum_i \Xi_i^{(\beta)}(t_c) \right|.$$

Table 6: Available integrators for solver comparison.

Integrator	Code	Order	Symplectic
Velocity Verlet	`vv`	2	Yes
Leapfrog	`lf`	2	Yes
Runge-Kutta 4	`rk4`	4	No
PEFRL	`pefrl`	4	Yes
Yoshida 6	`y6`	6	Yes
Adams-Bashforth	`ab4`	4	No

First Law of Identity provides a solver-independent invariant: $\delta\Xi_{\alpha\beta}$ grows monotonically for non-symplectic solvers due to artificial information injection, while symplectic solvers maintain bounded $\delta\Xi$. A symplectic solver showing unbounded $\delta\Xi$ growth indicates a force-field bug, not a solver defect.

```
[solver_compare]
enabled             = true
solvers             = ["vv","pefrl"]
checkpoint_interval = 1000
compare_metrics     = ["pos_rmsd","energy_drift","info_divergence"]
output_path         = "solver_cmp.csv"
```

Listing 5: Solver comparison configuration block.

19. Slimmed-Down Runtime Block

Full UIISCF bookkeeping has a cost. The per-particle Ξ ODE adds one float per particle per step; the sparse conductance matrix adds memory proportional to the number of proximate pairs; the event log grows with simulation time. For large-scale production runs — millions of atoms, millions of steps — this cost may be unacceptable.

The slim mode addresses this by suspending all information-theoretic bookkeeping while preserving the classical conservation checks. The fundamental invariants (energy, momentum, charge, particle count) are still validated. The identity-information block is frozen: Ξ_i and $\mho_{ij}$ are not updated. Event logging is reduced to a summary. The result is a run that is indistinguishable from a classical MD run in terms of output, but which can be trivially upgraded to full UIISCF mode by changing one configuration line.

Table 7: Feature availability in full vs. slim runtime modes.

Feature	Full UIISCF	Slim mode
Positions, velocities, forces	✓	✓
Kinetic / potential energy	✓	✓
Thermostat / barostat	✓	✓
Per-atom Ξ_i tracking	✓	—
Conductance matrix $\mho_{ij}$	✓	—
Scale quota $\mathcal{Q}_S$	✓	Aggregate only
Event log (`.evlog`)	✓	Summary only
Subatomic identity matrices	✓	—
Solver comparison	✓	—
Dark-sector projection	✓	—
Detector reconstruction	✓	—

20. Version Constraints

The UIISCF is implemented in stages, gated by version. This is a deliberate engineering decision: delivering the full formalism at once would require users to adopt the entire information-tracking infrastructure before they have any practical need for it. Version gating allows the formalism to be introduced incrementally, with each version boundary corresponding to a tested, validated capability plateau.

Version 5.1.4 is the classical MD baseline. Everything up to and including the matrix-force dispatch table, event logging, and formation-defect tracking is available without any subatomic content. This version is suitable for the vast majority of chemistry and materials-science simulations.

Version 5.1.5 activates the subatomic layer: quarks, leptons, bosons, information tracking, annihilation, and the full UIISCF state vector. It requires explicit opt-in for every subatomic feature and will reject configurations that enable subatomic channels without the information tracking infrastructure.

Table 8: Feature version gate table.

Feature	Min. version	Notes
Classical MD, matrix-force policy	v5.1.4	Baseline
Event logging (`.evlog`)	v5.1.4	Bond, formation, defect events
Formation-defect bridge	v5.1.4	Formation trajectory format required
Per-atom Ξ_i tracking	v5.1.5	`info_track = true`
Conductance matrix $\mho_{ij}$	v5.1.5	Sparse; max 10^6 non-zeros
Scale identity ladder (subatomic)	v5.1.5	`subatomic_mode = true`
Quark/lepton identity matrices	v5.1.5	Requires `full_qcd = false`
Boson identity matrix	v5.1.5	Photon proxy + weak proxy
Annihilation events	v5.1.5	Lepton and hadron channels
Solver comparison state	v5.1.5	Any two registered integrators
Dark-sector projection	v5.1.5	Phenomenological only
Detector projection/reconstruction	v5.1.5	Synthetic; no hardware interface
Proper-time correction	v5.2	*Planned*
Explicit electron tracking	v5.2	*Planned*
Full relativistic boost	v5.2	*Planned*

20.1 Activation Checklist (v5.1.5)

Before enabling any subatomic feature, verify:

1. `subatomic_mode = true` in the kernel block.
2. `info_track = true` in the run or kernel block.
3. `full_qcd = false` in any quark or boson block (no default).
4. Kernel version declared at `"5.1.5"` or higher.
5. Event log path configured.

Second Law of Identity applies at the version level: a configuration with $m \geq 3$ concurrent v5.1.5 feature blocks enabled and all guard conditions satisfied will automatically activate the full UIISCF state. A partial configuration raises `ERR_PARTIAL_UIISCF` and is rejected. Partial activation is not supported; the formalism is all-or-nothing at v5.1.5.

---

21. Conservation Summary and Global Invariants

Table 9: Global invariants in UIISCF.

Quantity	Symbol	Conservation type	Violation handler
Total energy	E_{tot}	Approximate (NVE)	Rescale thermostat
Total momentum	$\mathbf{P}$	Exact	Fatal error
Total charge	Q_{tot}	Exact	Fatal error
Particle count	N	Exact (non-annihil)	Fatal error
Lepton number	L_{tot}	Exact	Fatal error
Baryon number	B_{tot}	Exact	Fatal error
Total Ξ	$\sum_i \Xi_i$	Monotone non-increasing	Logged; not fatal
Entropy	S_{tot}	Monotone non-decreasing	Logged; not fatal
Scale quota sum	$\sum_S \mathcal{Q}_S$	Bounded change per step	Warning if $\lvert\Delta\mathcal{Q}_S\rvert > 10$

The information invariants (rows 7–8) are soft: they are tracked and logged but do not cause fatal errors. This is intentional. The First Law of Identity is a physical law, not an implementation constraint, and apparent violations indicate interesting physics rather than bugs. Only a spontaneous *increase* in $\sum_i \Xi_i$ without a corresponding event is treated as a bug (`ERR_INFO_CREATED`).

22. Complete Reference Script

A complete annotated configuration script demonstrating all UIISCF features:

```
[meta]
version         = "5.1.5"
script_id       = "UIISCF-REF-001"
description     = "Full UIISCF feature demonstration"

[material]
file            = "system.xyz"
format          = "xyz"

[run]
integrator      = "velocity_verlet"
dt              = 0.5
steps           = 500000
thermostat      = "nose_hoover"
thermostat_T    = 300.0
barostat        = "parrinello_rahman"
barostat_P      = 1.0
info_track      = true
info_dissip     = 0.001

[kernel]
mode                 = "full"
subatomic_mode       = true
event_rate_density   = 0.01
event_energy_guard   = 5.0
event_queue_depth    = 1000
```

```
event_log_path       = "run.evlog"
energy_tol           = 1.0e-4
momentum_tol         = 1.0e-6

[quark]
enabled          = true
flavours         = ["u","d","s"]
alpha_s          = 0.18
kappa_string     = 0.18
full_qcd         = false          # MANDATORY
cai_channel      = true

[boson]
enabled          = true
photon_proxy     = true
weak_proxy       = false
full_qcd         = false          # MANDATORY

[dark_sector]
enabled          = false

[solver_compare]
enabled              = true
solvers              = ["vv","pefrl"]
checkpoint_interval = 5000
output_path          = "solver_cmp.csv"

[observe]
interval         = 100
properties       =
   ["temperature","pressure","energy","Xi_total","quota_S"]
output           = "observables.chart"

[export]
trajectory           = "run.xyza"
formation            = "run.xyzf"
checkpoint           = "run.xyzc"
checkpoint_interval = 10000
reco_output          = "run.reco"
```

Listing 6: Complete UIISCF reference configuration (v5.1.5).

Summary of the Two Laws

First Law of Identity — Conservation of Distinguishability (canonical form).

$$\mathscr{P}_i \leftrightarrow \mathscr{P}_j \implies \Delta\Xi_{ij} < 0 \ \text{ or } \ \Delta S_{ij} > 0.$$

No physical interaction preserves or increases total recoverable distinguishability. Information is either converted to entropy or lost to the environment. This law holds at all scales and is scale-independent. It is the reason detector reconstruction is a lossy process; it is the reason annihilation destroys identity irreversibly; it is the reason that every interaction in any simulation must reduce the information budget of at least one participant.

Second Law of Identity — Scale-Coalescence Birth Criterion (canonical form).

$$m \geq 3, \quad d_{ij}^{(n)} \leq \ell_{\text{bond}}^{(n)}, \quad B_n(C_n) \geq \tau_B \implies \text{Exist}\Big(\mathscr{P}_*^{(n+1)}\Big) = 1, \quad \Delta\mathcal{Q}_S = +1.$$

Complexity emerges through coalescence. Every stable higher-scale identity requires at minimum three lower-scale identities satisfying proximity and bond maturity. The new identity inherits a reduced but positive information budget, consistent with the First Law. Quarks become hadrons; nucleons become nuclei; atoms become molecules. The universe does this at every scale, continuously, and has done so for 1.4×10^{10} years before anyone wrote it down.

What CERN produces in abundance is not particles but information about interactions.
Working backwards from that information to a coherent picture of identity is the whole problem.
Theoretical Division | UIISCF Rev. 1.0 | May 21, 2026